SAURABH SHINDE

AI Engineer | LLMs, RAG & Agentic AI
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PROFESSIONAL SUMMARY

AI Engineer with **4+ years** of experience building, evaluating and deploying **LLM, RAG and agentic AI** applications, alongside classical ML and MLOps pipelines. Delivered governed LLM workstreams aligned with the **EU AI Act**, built guardrails and content-moderation logic, and shipped Python services on AWS and Azure with Docker and CI/CD. Experienced in prompt engineering, vector search, LLM evaluation (RAGAS, LLM-as-judge), observability and responsible AI in regulated environments. **MSc in Artificial Intelligence** and **AWS Certified Machine Learning Engineer, Associate**.

EDUCATION

MSc Artificial Intelligence | Dublin Business School, Dublin, Ireland — **Jan 2025 – May 2026**
- Dissertation: ShastraShaw, a production multilingual legal RAG system (English, Hindi, Marathi).

B.E. Computer Engineering | Nashik, India — **Aug 2018 – Jul 2022**

PROFESSIONAL EXPERIENCE

AI Engineer (Part-time) | Orcawise Limited, Dublin, Ireland (Remote) — **May 2025 – Present**
- Delivered governed LLM workstreams across a **12-month** responsible AI program, building **15+ prompt and guardrail configurations** aligned with EU AI Act requirements.
- Implemented safety refusals and content moderation logic that reduced non-compliant model outputs by **~30%** across tested use cases.
- Integrated LLMs, RAG and search into RAIMS APIs, improving retrieval precision by **~20%** on internal compliance queries.
- Optimized RAG data pipelines and RAIMS chat performance, cutting average response latency by **~25%** for a multi-user platform.
- Collaborated in a mentor-led agile team of **8+ engineers**, shipping through Jira and GitLab in weekly sprints.

  Tech: LLMs, RAG, Guardrails, Prompt Engineering, REST APIs, EU AI Act, Jira, GitLab

AI Engineer | Capgemini, Navi Mumbai, India — **Dec 2022 – Dec 2024**
- Developed and deployed production LLM and RAG solutions that automated document-analysis workflows, integrating with enterprise systems and cutting manual workloads by **20%**.
- Applied prompt engineering and structured output validation to improve the reliability and consistency of LLM outputs in production.
- Engineered end-to-end MLOps pipelines (Docker, DVC, MLflow, GitHub Actions CI/CD), reducing model deployment time by **30%** across compliance-focused financial-services projects.
- Built production model monitoring with automated drift detection, fairness evaluation and explainability (SHAP, LIME), applying responsible-AI and data-privacy practices.
- Built and deployed containerised Python APIs and LLM-enabled workflows on AWS (SageMaker, Lambda, S3, ECS).
- Applied BERT and RoBERTa to document classification, improving accuracy by **15%** and reducing training time by **25%** with mixed-precision training.
- Worked with technical and non-technical stakeholders to turn regulated-domain requirements into maintainable, testable AI solutions.

  Tech: Python, LLMs, RAG, BERT / RoBERTa, Docker, DVC, MLflow, GitHub Actions, AWS (SageMaker, Lambda, S3, ECS), SHAP, LIME

Junior AI Engineer | Qriocity, Chennai, India — **Dec 2021 – Dec 2022**
- Built Python classification and regression models (scikit-learn, pandas) for an interactive learning platform.
- Ran the full ML workflow: data cleaning, feature engineering, training, hyperparameter tuning and evaluation.
- Used cross-validation, precision, recall and F1-score to compare models and reduce overfitting.
- Worked with product and curriculum teams to translate user requirements into ML-driven features.

  Tech: Python, scikit-learn, pandas

KEY PROJECTS

ShastraShaw | Multilingual Legal RAG System Live Site — **MSc Dissertation**
- **Eliminated hallucinated citations** in a production legal-RAG system, raising automated faithfulness from **0.33 to 0.88** by building a multi-layer guardrail stack (anti-hallucination prompting, reranking, citation verification with auto-retry).
- Lifted retrieval accuracy to a perfect **100% Hit@5 (from 93%)** across a 4-domain legal benchmark by adding a cross-encoder reranking layer on top of hybrid BM25 + FAISS retrieval.
- Built an LLMOps evaluation pipeline (RAGAS + lawyer-reviewed benchmarking) that **more than tripled context precision (0.21 to 0.73)**, catching citation and legal-provision errors before deployment.

- Operated the system as **production infrastructure, not a prototype**: AWS (EC2, S3, ECR), CloudWatch logging and alerting, rate-limiting, automated backups, Firebase auth.
- Built a multilingual (English, Hindi, Marathi) legal assistant delivering grounded, actionable guidance across 4 legal domains, expanding access to justice for **self-represented citizens**.

  Tech: GPT-4o, FAISS, BM25, Cross-Encoder Reranking, InLegalBERT, MarianMT, RAGAS, FastAPI, Docker, AWS (EC2, S3, CloudWatch), Firebase

AriseAI | Agentic RAG Assistant [Live Demo] Apr 2025 – 2026

- Built a LangGraph application covering query triage, retrieval, document grading, query reformulation and grounded answer generation over a FAISS vector store.
- Added Pydantic structured outputs, citation verification and confidence-based fallbacks to reduce unsupported responses.
- Created an LLM-as-judge evaluation harness measuring retrieval accuracy, faithfulness and relevance across a 30-question benchmark.
- Cut LLM API calls per query from **3-4 to 2**, added a **19-case** Pytest regression suite and deployed with Docker.

  Tech: LangGraph, Gemini, FAISS, Pydantic, Pytest, Docker

Safe Waters | Agentic Advisory Pipeline

TechIreland National AI Challenge 2026 ***Community Award Winner***

- Orchestrated an agentic pipeline combining live environmental API data, model-based forecasting and rule-based guardrails, moving monitoring of **10 beaches** from daily manual review to exception-only alerts.
- Integrated GPT-4o tool-calling with retrieval-augmented context, grounding advisories in real-time regulatory data and prior human-reviewer feedback.
- Designed auditable, human-in-the-loop escalation guardrails and validated the logic with automated tests.

  Tech: GPT-4o, Tool Calling, RAG, API Integration

SmartAQ | Forecasting with MLOps Pipeline Jul – Oct 2025

- Built LSTM and LightGBM forecasting models, achieving a **15% RMSE reduction** against the project baseline.
- Created a reproducible DVC, Docker and GitHub Actions pipeline; deployed batch and real-time inference on AWS SageMaker with Step Functions and CloudWatch.

  Tech: LSTM, LightGBM, AWS (SageMaker, Step Functions, CloudWatch), DVC, Docker, GitHub Actions

CERTIFICATIONS

- AWS Certified Machine Learning Engineer, Associate (Nov 2025)
- Master's Programme in AI Engineering, Simplilearn x IBM (Aug 2022)
- MLOps Foundations, iNeuron | Python for Data Science, IBM

TECHNICAL SKILLS

Languages: Python, SQL

LLMs & Agentic AI: LangGraph, LangChain, AI Agents, Tool Use / Function Calling, Prompt Engineering, Model Context Protocol (MCP), OpenAI / Azure OpenAI, Claude API, Gemini, AWS Bedrock, LLM Fine-tuning

RAG & Retrieval: RAG, Vector Databases (FAISS), Embeddings, Hybrid Search, Reranking, Semantic Search

LLM Evaluation & Responsible AI: LLM-as-Judge, RAGAS, Hallucination Mitigation, Guardrails, EU AI Act, AI Governance, Explainability (SHAP, LIME)

MLOps & Cloud: Docker, Kubernetes (basics), CI/CD (GitHub Actions, GitLab), MLflow, DVC, Airflow, AWS (SageMaker, Lambda, S3, ECS, CloudWatch), Azure

Backend & ML: FastAPI, Flask, REST APIs, Pydantic, PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, Pandas